QA Test Cases

medrio.xmltopdf · .NET 8 Upgrade & Security Hardening Validation

Scope: Patient Data Listings PDF Generation (UI-driven, no API access) | **Total Test Cases: 32**

Test Coverage Summary

Module	Total TCs	Critical	High	Medium
End-to-End PDF Generation	**8**	**4**	**4**	**–**
PDF Content & Layout Regression	**6**	**4**	**2**	**–**
Security – XXE Prevention (Optional / Dev Dependency)	**2**	**2**	**–**	**–**
Security – Error Message Leakage	**2**	**–**	**2**	**–**
Security – Log Injection Mitigation	**2**	**–**	**2**	**–**
Performance & Stability	**4**	**–**	**4**	**–**
UI Smoke	**8**	**4**	**4**	**–**
TOTAL	**32**	**14**	**18**	**0**

Priority Legend:

Critical	High	Medium

End-to-End PDF Generation

TC-E2E-001 | Small Dataset – Patient Data Listing Generation & PDF Download (Submission Casebooks)	
Module	End-to-End PDF Generation
Priority	**Critical**
Test Type	Functional
Preconditions	1. medrio.xmltopdf service deployed with .NET 8 build. 2. medrio.edc environment is accessible. 3. A study with ≤ 50 patients and ≤ 5 visits is available. 4. User has QA/Sponsor-level role with listing generation rights.
Test Steps	1. Log into EDC application using valid credentials. 2. Navigate to the Analysis >> Submission Casebooks for the small-dataset study. 3. Select the Sites, Subjects and Forms configuration (no custom filters). 4. Click ‘Download’ and note the time stamp. 5. Monitor the Job Queue (In EDC Admin) until the job shows status = Successful. 6. Click on 'Click here to Download' link. 7. Open the downloaded PDF and inspect every page.
Expected Result	✓ Job status transitions: Processing → Successful. No 'Failed' or 'Error' status. ✓ PDF file is downloadable and opens without corruption warnings. ✓ Page headers contain subject identifier, medrio identifier, site and group. ✓ Page footers contain timestamp, study name and page numbers in format 'X of Y'. ✓ All patient rows are present. No row duplication or missing rows. ✓ Column headers match the configured listing definition. ✓ No blank pages, no truncated text, no overlapping text blocks.
Notes	Baseline test. Compare PDF layout with a pre-upgrade snapshot if available.

Result	

TC-E2E-002 | Small Dataset – Patient Data Listing Generation & PDF Download (eCRF Screenshots)	
Module	End-to-End PDF Generation
Priority	**Critical**
Test Type	Functional
Preconditions	1. medrio.xmltopdf service deployed with .NET 8 build. 2. medrio.edc environment is accessible. 3. A study with ≤ 50 patients and ≤ 5 visits is available. 4. User has QA/Sponsor-level role with listing generation rights.
Test Steps	1. Log into EDC application using valid credentials. 2. Navigate to the Analysis >> eCRF Screenshots for the small-dataset study. 3. Select the Sites, Subjects and Forms configuration (no custom filters). 4. Click 'Download' and note the time stamp. 5. Monitor the Job Queue (In EDC Admin) until the job shows status = Successful. 6. Click on 'Click here to Download' link. 7. Open the downloaded PDF and inspect every page.
Expected Result	✓ Job status transitions: Processing → Successful. No 'Failed' or 'Error' status. ✓ PDF file is downloadable and opens without corruption warnings. ✓ Page headers contain subject identifier, medrio identifier, site, visit, group and form. ✓ Page footers contain timestamp, study name and page numbers in format 'X'. ✓ All patient rows are present. No row duplication or missing rows. ✓ Column headers match the configured listing definition. ✓ No blank pages, no truncated text, no overlapping text blocks.
Notes	Baseline test. Compare PDF layout with a pre-upgrade snapshot if available.
Result	

TC-E2E-003 | Large / Complex Dataset – High-Volume PDF Generation (Submission Casebooks)	
Module	End-to-End PDF Generation
Priority	**Critical**
Test Type	Functional
Preconditions	1. medrio.xmltopdf service deployed with .NET 8 build. 2. A study with ≥ 500 patients, multiple visits, and wide data columns is available. 3. User has listing generation rights.
Test Steps	1. Log into EDC application using valid credentials. 2. Navigate to the Analysis >> Submission Casebooks for the small-dataset study. 3. Select the Sites, Subjects and Forms configuration (no custom filters). 4. Click 'Download' and note the time stamp. 5. Monitor the Job Queue (In EDC Admin) until the job shows status = Successful. 6. Click on 'Click here to Download' link. 7. Open the downloaded PDF and inspect every page.
Expected Result	✓ Job completes without timeout errors. ✓ Generation time is within ±20% of the pre-upgrade baseline (or within team's acceptable SLA). ✓ PDF page count matches expected total (verify against known pre-upgrade count if available). ✓ All pages render correctly – no truncated rows at page breaks. ✓ Table column widths are consistent across all pages. ✓ No 'OutOfMemoryException' or 'System.Threading' errors in the job log.
Notes	Note actual start and end timestamps for performance comparison in TC-PERF-001.

TC-E2E-004 | Large / Complex Dataset – High-Volume PDF Generation (eCRF Screenshots)	
Module	End-to-End PDF Generation
Priority	**Critical**
Test Type	Functional
Preconditions	1. medrio.xmltopdf service deployed with .NET 8 build. 2. A study with ≥ 500 patients, multiple visits, and wide data columns is available. 3. User has listing generation rights.
Test Steps	1. Log into EDC application using valid credentials. 2. Navigate to the Analysis >> eCRF Screenshots for the small-dataset study. 3. Select the Sites, Subjects and Forms configuration (no custom filters). 4. Click ‘Download’ and note the time stamp. 5. Monitor the Job Queue (In EDC Admin) until the job shows status = Successful. 6. Click on 'Click here to Download' link. 7. Open the downloaded PDF and inspect every page.
Expected Result	✓ Job completes without timeout errors. ✓ Generation time is within ±20% of the pre-upgrade baseline (or within team's acceptable SLA). ✓ PDF page count matches expected total (verify against known pre-upgrade count if available). ✓ All pages render correctly – no truncated rows at page breaks. ✓ Table column widths are consistent across all pages. ✓ No 'OutOfMemoryException' or 'System.Threading' errors in the job log.
Notes	Note actual start and end timestamps for performance comparison in TC-PERF-001.
Result	

TC-E2E-005 | Filter Variations (Submission Casebooks)	
Module	End-to-End PDF Generation
Priority	**High**
Test Type	Functional
Preconditions	1. Filters must be decided
Test Steps	1. Log into EDC application using valid credentials. 2. Navigate to the Analysis >> Submission Casebooks for the small-dataset study. 3. Select the Sites, Subjects and Forms configuration with custom filters. 4. Click ‘Download’ and note the time stamp. 5. Monitor the Job Queue (In EDC Admin) until the job shows status = Successful. 6. Click on 'Click here to Download' link. 7. Open the downloaded PDF and inspect every page.
Expected Result	✓ PDF contains only subjects within the specified filter. ✓ Column order and sorting are consistent with how configs were defined. ✓ Neither PDF shows data from the wrong configuration (no cross-contamination).
Notes	Critical to rule out caching or serialization regressions introduced by the .NET 8 upgrade.
Result	

TC-E2E-006 | Filter Variations (eCRF Screenshots)	
Module	End-to-End PDF Generation
Priority	**High**
Test Type	Functional
Preconditions	1. Filters must be decided

Test Steps	1. Log into EDC application using valid credentials. 2. Navigate to the Analysis >> Submission Casebooks for the small-dataset study. 3. Select the Sites, Subjects and Forms configuration with custom filters. 4. Click 'Download' and note the time stamp. 5. Monitor the Job Queue (In EDC Admin) until the job shows status = Successful. 6. Click on 'Click here to Download' link. 7. Open the downloaded PDF and inspect every page.
Expected Result	✓ PDF contains only subjects within the specified filter. ✓ Column order and sorting are consistent with how configs were defined. ✓ Neither PDF shows data from the wrong configuration (no cross-contamination).
Notes	Critical to rule out caching or serialization regressions introduced by the .NET 8 upgrade.
Result	

TC-E2E-007 | Concurrent Job Execution – Two Simultaneous Listing Requests	
Module	End-to-End PDF Generation
Priority	**High**
Test Type	Functional
Preconditions	1. Two separate studies with different patient volumes are available. 2. Two browser sessions (or two user accounts) with listing generation rights.
Test Steps	1. In Browser Session 1, navigate to Study A listing generation page. 2. In Browser Session 2, navigate to Study B listing generation page. 3. Trigger generation in Session 1 first, then immediately trigger generation in Session 2. 4. Monitor both job queues simultaneously until both reach Completed status. 5. Download and inspect both PDFs.
Expected Result	✓ Both jobs complete successfully without one blocking or killing the other. ✓ Study A PDF contains only Study A data; Study B PDF contains only Study B data. ✓ No data mixing between concurrent jobs. ✓ Job queue shows correct statuses for each job independently.
Notes	Validates thread safety after .NET 8 runtime upgrade; critical for multi-tenant stability.
Result	

TC-E2E-008 | Job Failure – Graceful Error Handling and Recovery (Optional)	
Module	End-to-End PDF Generation
Priority	**High**
Test Type	Negative / Error Handling
Preconditions	1. Dev team has identified or can simulate a configuration that causes XmlToPdf to return a controlled failure. 2. OR a study with no patients (empty dataset) can be used to trigger a handled error.
Test Steps	1. Navigate to Patient Data Listings for the failure-inducing study or config. 2. Trigger listing generation. 3. Wait for the job to reach its terminal status (expect 'Failed'). 4. Observe the error message displayed in the EDC UI. 5. Check that no partial PDF is available for download. 6. Immediately trigger a new listing for a known-good study to verify recovery.
Expected Result	✓ Job status shows 'Failed' (or equivalent) – not a hang or infinite 'In Progress'. ✓ Error message is user-friendly: e.g., 'Listing generation failed. Please retry or contact support.' No stack traces or internal paths exposed. ✓ No partial/corrupt PDF is offered for download. ✓ The subsequent listing for the known-good study completes successfully.

	✓ No other running jobs are disrupted by this failure.
Notes	Validates that error propagation from XmlToPdf to EDC JobProcessor is intact after the upgrade.
Result	

PDF Content & Layout Regression

TC-LAY-001 | Page Header & Footer Integrity Across All Pages (Submission Casebooks)	
Module	PDF Content & Layout Regression
Priority	**Critical**
Test Type	Regression
Preconditions	1. A multi-page PDF generated in TC-E2E-001 or TC-E2E-003 is available. 2. A reference/baseline PDF from the pre-upgrade build is available (if possible).
Test Steps	1. Open the generated PDF. 2. Inspect page header: Subject Identifier, Medrio ID, Site and Group 3. Inspect page footer: Time stamp, study name and page number like 'X of Y' 4. Navigate to a middle page (e.g., page 5 or 10) and inspect the same header fields. 5. Navigate to the last page and inspect header and footer. 6. Check footer on all inspected pages for page number format. 7. If baseline PDF is available, compare headers side-by-side with baseline.
Expected Result	✓ Study name, date are identical across all inspected pages. ✓ Footer shows 'Page X of Y' where Y is the total page count (consistent). ✓ No header or footer is missing on any page. ✓ Font, size, and alignment of header/footer match baseline (or prior experience). ✓ No header/footer overlaps with table content.
Result	

TC-LAY-002 | Page Header & Footer Integrity Across All Pages (eCRF Screenshots)	
Module	PDF Content & Layout Regression
Priority	**Critical**
Test Type	Regression
Preconditions	1. A multi-page PDF generated in TC-E2E-002 or TC-E2E-004 is available. 2. A reference/baseline PDF from the pre-upgrade build is available (if possible).
Test Steps	1. Open the generated PDF. 2. Inspect page header: Subject Identifier, Medrio ID, Site and Group 3. Inspect page footer: Time stamp, study name and page number like 'X of Y' 4. Navigate to a middle page (e.g., page 5 or 10) and inspect the same header fields. 5. Navigate to the last page and inspect header and footer. 6. Check footer on all inspected pages for page number format. 7. If baseline PDF is available, compare headers side-by-side with baseline.
Expected Result	✓ Study name and date are identical across all inspected pages. ✓ Footer shows 'Page X of Y' where Y is the total page count (consistent). ✓ No header or footer is missing on any page. ✓ Font, size, and alignment of header/footer match baseline (or prior experience). ✓ No header/footer overlaps with table content.
Result	

TC-LAY-003 | Data Row Completeness – No Missing or Duplicated Patient Records (Submission Casebooks)	
Module	PDF Content & Layout Regression
Priority	**Critical**
Test Type	Regression
Preconditions	1. Known study with exactly N patients (e.g., 25 patients, verified in EDC). 2. Listing configuration includes all subjects.
Test Steps	1. Note the exact patient count from the EDC subject list before generating. 2. Generate the Patient Data Listing for this study. 3. Download the PDF and count total patient rows across all pages. 4. Cross-check each patient's Subject ID in the PDF against the EDC subject list. 5. Check for any duplicate Subject ID rows in the PDF.
Expected Result	✓ Patient row count in PDF equals the exact count recorded from EDC. ✓ Every Subject ID from EDC appears in the PDF exactly once. ✓ No Subject ID appears more than once (no duplication). ✓ No 'phantom' Subject IDs appear that do not exist in EDC. ✓ Data in each row (values, visit names) matches the data visible in EDC for that subject.
Notes	Direct regression check against XML serialization changes introduced by the security hardening.
Result	

TC-LAY-004 | Data Row Completeness – No Missing or Duplicated Patient Records (eCRF Screenshots)	
Module	PDF Content & Layout Regression
Priority	**Critical**
Test Type	Regression
Preconditions	1. Known study with exactly N patients (e.g., 25 patients, verified in EDC). 2. Listing configuration includes all subjects.
Test Steps	1. Note the exact patient count from the EDC subject list before generating. 2. Generate the Patient Data Listing for this study. 3. Download the PDF and count total patient rows across all pages. 4. Cross-check each patient's Subject ID in the PDF against the EDC subject list. 5. Check for any duplicate Subject ID rows in the PDF.
Expected Result	✓ Patient row count in PDF equals the exact count recorded from EDC. ✓ Every Subject ID from EDC appears in the PDF exactly once. ✓ No Subject ID appears more than once (no duplication). ✓ No 'phantom' Subject IDs appear that do not exist in EDC. ✓ Data in each row (values, visit names) matches the data visible in EDC for that subject.
Notes	Direct regression check against XML serialization changes introduced by the security hardening.
Result	

TC-LAY-005 | Character Encoding – Special Characters and International Data (Submission Casebooks)	
Module	PDF Content & Layout Regression
Priority	**High**
Test Type	Regression
Preconditions	1. At least one patient or visit has data containing special characters: accented letters (é, ü, ñ), symbols (©, ±, °), or non-Latin characters if applicable.
Test Steps	1. Identify the subject(s) with special characters in EDC. 2. Generate the listing that includes these subjects.

	3. Download and open the PDF. 4. Locate the cells with special characters and inspect rendering.
Expected Result	✓ Accented characters (é, ü, ñ) render correctly – no replacement with '?' or boxes. ✓ Symbols (©, ±, °) are preserved as-is. ✓ No garbled characters (e.g., 'Ã©' instead of 'é') indicating encoding regression. ✓ The encoding behavior matches what was seen before the .NET 8 upgrade.
Notes	Encoding can be subtly broken by .NET 8 XML reader defaults. Verify carefully.
Result	

TC-LAY-006 | Character Encoding – Special Characters and International Data (eCRF Screenshots)	
Module	PDF Content & Layout Regression
Priority	**High**
Test Type	Regression
Preconditions	1. At least one patient or visit has data containing special characters: accented letters (é, ü, ñ), symbols (©, ±, °), or non-Latin characters if applicable.
Test Steps	1. Identify the subject(s) with special characters in EDC. 2. Generate the listing that includes these subjects. 3. Download and open the PDF. 4. Locate the cells with special characters and inspect rendering.
Expected Result	✓ Accented characters (é, ü, ñ) render correctly – no replacement with '?' or boxes. ✓ Symbols (©, ±, °) are preserved as-is. ✓ No garbled characters (e.g., 'Ã©' instead of 'é') indicating encoding regression. ✓ The encoding behavior matches what was seen before the .NET 8 upgrade.
Notes	Encoding can be subtly broken by .NET 8 XML reader defaults. Verify carefully.
Result	

Security – XXE Prevention

TC-SEC-001 | XXE Payload via Job Trigger – System Stability and No Data Leak	
Module	Security – XXE Prevention
Priority	**Critical**
Test Type	Security / Negative
Preconditions	1. Dev/Security team has prepared a test XML payload containing a DTD-based external entity (XXE payload). 2. A test harness or dev-accessible entry point for the XmlToPdf service is available. 3. QA has observer access to the job queue in EDC UI.
Test Steps	1. Coordinate with Dev: submit the XXE payload through the agreed test entry point. 2. Observe the job status in the EDC job queue. 3. After the job reaches a terminal status, attempt to download any PDF for that job. 4. Review any error message visible in the UI. 5. Request Dev to verify the service logs show a controlled rejection (not a crash). 6. After the test, trigger a normal small-dataset listing (from TC-E2E-001) to confirm recovery.
Expected Result	✓ Job reaches 'Failed' status, NOT 'Completed'. ✓ No PDF is generated or available for download. ✓ Error message in UI does not contain file paths, server names, or internal XML content. ✓ The XmlToPdf service remains running – the subsequent normal job completes successfully.

	✓ Dev confirms logs show: XML parsing rejected due to DTD/external entity, no external HTTP call was made.
Notes	QA role is observer + UI validation. Dev/Security owns the payload construction and log verification.
Result	

TC-SEC-002 | Normal Valid XML is Not Blocked by XXE Protection	
Module	Security – XXE Prevention
Priority	**Critical**
Test Type	Security / Regression
Preconditions	1. TC-SEC-001 has been executed (XXE protection is active). 2. A standard listing configuration with a small dataset is available.
Test Steps	1. Navigate to Patient Data Listings for the small-dataset study. 2. Select the standard listing configuration (same as TC-E2E-001). 3. Trigger listing generation. 4. Wait for job completion and download the PDF.
Expected Result	✓ Job completes with status 'Completed' – XXE protection does NOT block valid listings. ✓ PDF is generated, downloadable, and contains correct data. ✓ No false-positive 'security rejection' error appears. ✓ This result is identical to TC-E2E-001 outcome.
Notes	Confirms XXE fix is targeted – it does not over-block legitimate XML traffic.
Result	

Security – Error Message Leakage

TC-SEC-003 | Failed Job Error Messages Do Not Expose Internal Details (Submission Casebooks)	
Module	Security – Error Message Leakage
Priority	**High**
Test Type	Security / Negative
Preconditions	1. A configuration or study that causes a handled job failure is available (dev-simulated failure).
Test Steps	1. Trigger a listing generation that is expected to fail (use the same method as TC-E2E-005). 2. Wait for the 'Failed' status. 3. Carefully read and record the exact error message shown in the EDC UI. 4. Check if the message contains any of the following: file paths, server hostnames, stack traces, .NET exception names, XML content, internal class names.
Expected Result	✓ Error message is a generic, user-facing message only (e.g., 'Report generation failed. Please contact support with reference ID: XXXX'). ✓ No stack trace, no .NET exception type names, no internal file paths are visible. ✓ No XML fragment or partial payload is echoed back. ✓ Message is identical in structure to pre-upgrade error messages (no regression in error formatting).
Notes	Log injection mitigation may alter how errors are formatted. Validate that sanitized errors still surface meaningfully to users.
Result	

TC-SEC-004 | Failed Job Error Messages Do Not Expose Internal Details (eCRF Screenshots)

Module	Security – Error Message Leakage
Priority	**High**
Test Type	Security / Negative
Preconditions	2. A configuration or study that causes a handled job failure is available (dev-simulated failure).
Test Steps	5. Trigger a listing generation that is expected to fail (use the same method as TC-E2E-005). 6. Wait for the 'Failed' status. 7. Carefully read and record the exact error message shown in the EDC UI. 8. Check if the message contains any of the following: file paths, server hostnames, stack traces, .NET exception names, XML content, internal class names.
Expected Result	✓ Error message is a generic, user-facing message only (e.g., 'Report generation failed. Please contact support with reference ID: XXXX'). ✓ No stack trace, no .NET exception type names, no internal file paths are visible. ✓ No XML fragment or partial payload is echoed back. ✓ Message is identical in structure to pre-upgrade error messages (no regression in error formatting).
Notes	Log injection mitigation may alter how errors are formatted. Validate that sanitized errors still surface meaningfully to users.
Result	

Security – Log Injection Mitigation

TC-SEC-005 | Job With Unusual Study Name / Metadata Does Not Crash the Service (Submission Casebooks)	
Module	Security – Log Injection Mitigation
Priority	**High**
Test Type	Security / Negative
Preconditions	1. Dev team can create or has available a study whose name or metadata contains newline characters, CRLF sequences, or unusually long strings (≥ 500 characters) – to be tested in a non-production environment only. 2. QA observes via EDC job queue UI.
Test Steps	1. With dev support, trigger a listing job for the study with the unusual metadata. 2. Monitor the job queue for status. 3. If job is completed, download and inspect the PDF. 4. If job fails; note the error message. 5. After the test, trigger a normal job to verify service stability.
Expected Result	✓ Service does NOT crash or restart. ✓ Job either completes successfully or fails with a handled, user-friendly error. ✓ No 500-level error surfaces to the EDC UI. ✓ Normal job following this test completes successfully. ✓ Dev confirms logs show the unusual characters were sanitized, not raw-written.
Notes	QA is validating service stability and UI behavior only. Log content verification is owned by Dev.
Result	

TC-SEC-006 | Job With Unusual Study Name / Metadata Does Not Crash the Service (eCRF Screenshots)	
Module	Security – Log Injection Mitigation
Priority	**High**
Test Type	Security / Negative

Preconditions	1. Dev team can create or has available a study whose name or metadata contains newline characters, CRLF sequences, or unusually long strings (≥ 500 characters) – to be tested in a non-production environment only. 2. QA observes via EDC job queue UI.
Test Steps	1. With dev support, trigger a listing job for the study with the unusual metadata. 2. Monitor the job queue for status. 3. If job is completed, download and inspect the PDF. 4. If job fails; note the error message. 5. After the test, trigger a normal job to verify service stability.
Expected Result	✓ Service does NOT crash or restart. ✓ Job either completes successfully or fails with a handled, user-friendly error. ✓ No 500-level error surfaces to the EDC UI. ✓ Normal job following this test completes successfully. ✓ Dev confirms logs show the unusual characters were sanitized, not raw-written.
Notes	QA is validating service stability and UI behavior only. Log content verification is owned by Dev.
Result	

Performance & Stability

TC-PERF-001 | **Generation Time Sanity – Small vs Large Dataset Comparison (Submission Casebooks)**	
Module	Performance & Stability
Priority	**High**
Test Type	Performance
Preconditions	1. Small study (≤ 50 patients) and large study (≥ 500 patients) are available. 2. Pre-upgrade timing baseline is available (even as rough estimate from team's memory or historical logs).
Test Steps	1. Trigger listing for the small study; record time from click to 'Completed' status. 2. Trigger listing for the large study; record time from click to 'Completed' status. 3. Run each scenario twice to check for consistency. 4. Compare recorded times against baseline or team expectations.
Expected Result	✓ Small dataset completes in ≤ [team's agreed threshold – document actual baseline]. (or within team's known baseline). ✓ Large dataset completes in ≤ [team's agreed threshold – document actual baseline]. ✓ Second run of each is within ±15% of the first run (no cold-start instability). ✓ No job stalls at 'In Progress' for longer than 5× expected time.
Notes	Rough time measurements are sufficient. Exact profiling is owned by Dev/Infra.
Result	

TC-PERF-002 | **Generation Time Sanity – Small vs Large Dataset Comparison (eCRF Screenshots)**	
Module	Performance & Stability
Priority	**High**
Test Type	Performance
Preconditions	1. Small study (≤ 50 patients) and large study (≥ 500 patients) are available. 2. Pre-upgrade timing baseline is available (even as rough estimate from team's memory or historical logs).
Test Steps	1. Trigger listing for the small study; record time from click to 'Completed' status. 2. Trigger listing for the large study; record time from click to 'Completed' status.

	3. Run each scenario twice to check for consistency. 4. Compare recorded times against baseline or team expectations.
Expected Result	✓ Small dataset completes in ≤ [team's agreed threshold – document actual baseline]. (or within team's known baseline). ✓ Large dataset completes in ≤ [team's agreed threshold – document actual baseline]. ✓ Second run of each is within ±15% of the first run (no cold-start instability). ✓ No job stalls at 'In Progress' for longer than 5× expected time.
Notes	Rough time measurements are sufficient. Exact profiling is owned by Dev/Infra.
Result	

TC-PERF-003 | Stability Under Repeated Sequential Job Execution (8 Runs) (Submission Casebooks)	
Module	Performance & Stability
Priority	**High**
Test Type	Stability
Preconditions	1. At least two studies with different configurations are available. 2. Service has been deployed fresh for this testing cycle.
Test Steps	1. Trigger listing generation for Study A (Run 1). Wait for Completed. 2. Immediately trigger Study B (Run 2). Wait for Completed. 3. Alternate between Study A and Study B for 8 total runs. 4. Record pass/fail and approximate completion time for each run. 5. After all, 8 runs, download PDFs from Runs 1, 4, and 8 and spot-check content.
Expected Result	✓ All 8 runs complete with 'Completed' status. ✓ No progressive slowdown: Run 8 completion time should not be >30% slower than Run 1. ✓ No failures in Runs 5–8 (no build-up of instability). ✓ Spot-check PDFs (Runs 1, 4, 8) all contain correct, complete data. ✓ No service restart or memory-related error messages appear in the UI.
Notes	Detects memory leaks or connection pool exhaustion introduced by the .NET 8 migration.
Result	

TC-PERF-004 | Stability Under Repeated Sequential Job Execution (8 Runs) (eCRF Screenshots)	
Module	Performance & Stability
Priority	**High**
Test Type	Stability
Preconditions	1. At least two studies with different configurations are available. 2. Service has been deployed fresh for this testing cycle.
Test Steps	1. Trigger listing generation for Study A (Run 1). Wait for Completed. 2. Immediately trigger Study B (Run 2). Wait for Completed. 3. Alternate between Study A and Study B for 8 total runs. 4. Record pass/fail and approximate completion time for each run. 5. After all, 8 runs, download PDFs from Runs 1, 4, and 8 and spot-check content.
Expected Result	✓ All 8 runs complete with 'Completed' status. ✓ No progressive slowdown: Run 8 completion time should not be >30% slower than Run 1. ✓ No failures in Runs 5–8 (no build-up of instability). ✓ Spot-check PDFs (Runs 1, 4, 8) all contain correct, complete data. ✓ No service restart or memory-related error messages appear in the UI.
Notes	Detects memory leaks or connection pool exhaustion introduced by the .NET 8 migration.
Result	

UI Smoke

TC-UI-001 | Patient Data Listings Page Loads Without Errors (Submission Casebooks)	
Module	UI Smoke
Priority	**Critical**
Test Type	Smoke
Preconditions	1. Browser: Chrome (latest) and Firefox (latest). 2. Valid user session with listing access.
Test Steps	1. Open Chrome. 2. Log into EDC web application. 3. Navigate Analysis >> Submission Casebooks. 4. Observe the page load: check for console errors (F12 → Console tab). 5. Verify all UI controls are visible: study selector, configuration selector, Generate button. 6. Repeat in Firefox.
Expected Result	✓ Page loads within 5 seconds in both browsers. ✓ No JavaScript console errors of severity Error or Warning related to the listing page. ✓ Study selector is populated with available studies. ✓ Configuration selector updates correctly when a study is selected. ✓ 'Generate Listing' button is clickable (not greyed out).
Result	

TC-UI-002 | Patient Data Listings Page Loads Without Errors (eCRF Screenshots)	
Module	UI Smoke
Priority	**Critical**
Test Type	Smoke
Preconditions	1. Browser: Chrome (latest) and Firefox (latest). 2. Valid user session with listing access.
Test Steps	1. Open Chrome. 2. Log into EDC web application. 3. Navigate Analysis >> eCRF Screenshots. 4. Observe the page load: check for console errors (F12 → Console tab). 5. Verify all UI controls are visible: study selector, configuration selector, Generate button. 6. Repeat in Firefox.
Expected Result	✓ Page loads within 5 seconds in both browsers. ✓ No JavaScript console errors of severity Error or Warning related to the listing page. ✓ Study selector is populated with available studies. ✓ Configuration selector updates correctly when a study is selected. ✓ 'Generate Listing' button is clickable (not greyed out).
Result	

TC-UI-003 | Job Queue Status Updates Reflect Real-Time Job Progression (Submission Casebooks)	
Module	UI Smoke
Priority	**High**
Test Type	Smoke
Preconditions	1. A listing generation that takes > 10 seconds is expected (use large dataset).
Test Steps	1. Trigger listing generation for the large-dataset study.

	2. Immediately navigate to the Job Queue / job monitoring section. 3. Observe job status changes without refreshing manually (or refresh every 5 seconds if auto-update is not present). 4. Record each status observed: Processing → Successful.
Expected Result	✓ Job appears in the queue immediately after triggering (within 5 seconds). ✓ Status transitions are visible and logical (Processing → Successful). ✓ No phantom jobs appear (no jobs from other studies are shown in this user's queue unexpectedly). ✓ Download link appears only after status = Completed, not before.

TC-UI-004 | Job Queue Status Updates Reflect Real-Time Job Progression (eCRF Screenshots)	
Module	UI Smoke
Priority	**High**
Test Type	Smoke
Preconditions	1. A listing generation that takes > 10 seconds is expected (use large dataset).
Test Steps	1. Trigger listing generation for the large-dataset study. 2. Immediately navigate to the Job Queue / job monitoring section. 3. Observe job status changes without refreshing manually (or refresh every 5 seconds if auto-update is not present). 4. Record each status observed: Processing → Successful.
Expected Result	✓ Job appears in the queue immediately after triggering (within 5 seconds). ✓ Status transitions are visible and logical (Processing → Successful). ✓ No phantom jobs appear (no jobs from other studies are shown in this user's queue unexpectedly). ✓ Download link appears only after status = Completed, not before.
Result	

TC-UI-005 | PDF Download Link – Correct File Downloaded (Submission Casebooks)	
Module	UI Smoke
Priority	**Critical**
Test Type	Smoke
Preconditions	1. A successfully completed listing job is available. 2. Browser's download settings save files to a known location.
Test Steps	1. Locate a completed listing job in the job queue. 2. Click the 'Download PDF' link. 3. Check: (a) the download starts immediately, (b) file size > 0 KB, (c) filename reflects the study/config. 4. Open the downloaded file in a PDF viewer. 5. Check page 1 for correct study name and listing title.
Expected Result	✓ Download initiates within 3 seconds of clicking. ✓ Downloaded file is not 0 KB and is not corrupt. ✓ Filename pattern matches expected format (e.g., StudyName_PatientDataListing_YYYYMMDD.pdf). ✓ PDF opens without 'File is damaged or cannot be repaired' errors. ✓ Page 1 shows the correct study name confirming it is not a stale cached file.
Result	

TC-UI-006 | PDF Download Link – Correct File Downloaded (eCRF Screenshots)	
Module	UI Smoke
Priority	**Critical**

Test Type	Smoke
Preconditions	1. A successfully completed listing job is available. 2. Browser's download settings save files to a known location.
Test Steps	1. Locate a completed listing job in the job queue. 2. Click the 'Download PDF' link. 3. Check: (a) the download starts immediately, (b) file size > 0 KB, (c) filename reflects the study/config. 4. Open the downloaded file in a PDF viewer. 5. Check page 1 for correct study name and listing title.
Expected Result	✓ Download initiates within 3 seconds of clicking. ✓ Downloaded file is not 0 KB and is not corrupt. ✓ Filename pattern matches expected format (e.g., StudyName_PatientDataListing_YYYYMMDD.pdf). ✓ PDF opens without 'File is damaged or cannot be repaired' errors. ✓ Page 1 shows the correct study name confirming it is not a stale cached file.
Result	

TC-UI-007 | No New Unintended Error Messages Appear on Listing Pages (Submission Casebooks)	
Module	UI Smoke
Priority	**High**
Test Type	Smoke
Preconditions	1. All prior E2E tests have been executed.
Test Steps	1. Navigate through: study selection → config selection → generate → job queue → download PDF. 2. At each step, observe for any new banner messages, modal dialogs, or inline error text that were not present before the upgrade. 3. Attempt to re-download an already-downloaded PDF. 4. Navigate away and come back to the listing page.
Expected Result	✓ No new error banners appear during the normal happy path. ✓ Re-downloading a completed PDF works correctly the second time. ✓ Navigating away and returning to the page does not show a stale 'In Progress' status for a completed job. ✓ No '500 Internal Server Error' page is shown at any point during normal use.
Result	

TC-UI-008 | No New Unintended Error Messages Appear on Listing Pages (eCRF Screenshots)	
Module	UI Smoke
Priority	**High**
Test Type	Smoke
Preconditions	1. All prior E2E tests have been executed.
Test Steps	1. Navigate through: study selection → config selection → generate → job queue → download PDF. 2. At each step, observe for any new banner messages, modal dialogs, or inline error text that were not present before the upgrade. 3. Attempt to re-download an already-downloaded PDF. 4. Navigate away and come back to the listing page.
Expected Result	✓ No new error banners appear during the normal happy path. ✓ Re-downloading a completed PDF works correctly the second time. ✓ Navigating away and returning to the page does not show a stale 'In Progress' status for a completed job. ✓ No '500 Internal Server Error' page is shown at any point during normal use.
Result	